Year 8 Algebra Test

(Marks: /36)

(Working Out: /10)

Question 1 (3)

Copy and complete the following table. Use the rule given to work out what the OUT numbers on the function machine would be.

Add three to each IN number.

IN	OUT
3	6
-6	-3
14	17
24	27
-9	-6
20	23

Question 2 (3)

Copy and complete the following table. Use the rule given to work out what the OUT numbers on the function machine would be.

Take four from each IN number.

IN	OUT
5	1
-2	-6
18	14
29	25
-8	-12
13	17

Question 3 (3)

Copy and complete the following table. Use the rule given to work out what the OUT numbers on the function machine would be.

Add six to each IN number.

IN	OUT
1	7
-5	1
14	20
22	28
-4	0
11	17

Question 4 (3)

The famous scientist Alfred Finestein invented a machine that could change the mass of objects. Alfred set his machine to increase the mass of a chocolate cake three times and then decrease the mass of the cake by 100 grams. If the chocolate cake started at 450 g, what was its mass after it had been through Alfred's machine? 1250g

1

450
3
1350 - 100 = 1250

Question 5 (2)

If the average weight in kilograms of members of your class is *w*, write an expression for:

(a) The weight of someone who is 3 kg above average. w+3

(b) The weight of someone who is twice the average. 2w

Question 6 (2)

I start with a number and double it and then add 3 to it. If the number is represented by *n* then write an expression to show this situation. n2+3

Question 7 (2)

In the expression $5y - 2xy + 6 + 7x$:

(a) What is the coefficient of x? 7

(b) What is the constant term? 6

Question 8 (2)

Substitute $x = 3$ and $y = 1$ into the following and evaluate:

(a) $4x + 2y = 12 + 2 = 14$

(b) $6xy - 7 = 18 - 7 = 11$

Question 9 (2)

Substitute $x = 3$ and $y = 1$ into the following and evaluate:

(a) $2x + 8y = 6 + 8 = 14$

(b) $5xy - 2 = 15 - 2 = 13$

Question 10 (2)

Substitute $x = 2$ and $y = 4$ into the following and evaluate:

(a) $\frac{8x}{y} + 2x$ $= \frac{16}{4} + 32 = 4 + 32 = 36$

(b) $\frac{12}{x} + \frac{10xy}{2}$ $= \frac{12}{2} + \frac{80}{2} = 6 + 40 = 46$

Question 11 (3)

Simplify if possible:

(a) $9x - 3x$ $= 3x - x = 2x$

(b) $3b - 4a$ $= 3b - 4a$

(c) $4n + 28m - n + 12m$ $= 3n + 40m$

Question 12 (3)

Simplify:

(a) $10a \times 4b$ $= 5a \times 2b = 10ab$

(b) $2pq \times 6 \times r$ $pq \times 3 \times r = 3pqr$

(c) $-3m \times -n \times 4$ $= -3mn \times 4 = 12mn$

Question 13 (3)

Expand:

(a) $3(4 + x)$ $= 12+3x$

(b) $x(5y + 1)$ $= 5xy+x$

(c) $3x(4y + 10)$ $= 12xy+30x$

Question 14 (3)

Simplify by expanding and collecting like terms:

(a) $5(a + 2) + 3a$ $= 5a+10+3a = 8a+10$

(b) $11a(3 + 4b) + 4(2a - ab)$ $= 33a+44ab+8a-4ab = 33a+8a+44ab-4ab = 41a+40ab$

(c) $a(4b + 8c) + 2a(3b - 3c)$ $= 4ab+8ac+6ab-6ac = 4ab+6ab+2ac = 10ab+2ac$